

Electronically stimulated desorption of atoms from surfaces

Foreign atoms can be adsorbed on the surface of a crystalline semiconductor. For example, hydrogen atoms can be adsorbed on a silicon crystal's surface. Irradiations with high-energy electrons, photons and even ions are observed to induce the expulsion of such adsorbed atoms. Observations support the notion that desorption from crystals is fundamentally different from that from small molecules where electronic excitations are necessarily localized (Jennison and Emin, 1983a).

Irradiation of an ideal periodic surface can produce delocalized surface electronic excitations. The time that such an excitation can dwell in the vicinity of an adsorbed atom is $\sim \hbar/W_e$, where W_e is the electronic excitation's bandwidth. This time is usually much less than the time required to significantly displace an atom $\sim 1/\omega$, where ω is the relevant atomic-vibration frequency. Thus, delocalized electronic excitations are generally unable to displace adsorbed atoms and thereby cleave their bonds to surface atoms.

Electronically stimulated desorption of an adsorbed atom is therefore associated with the severe localization of an electronic surface excitation which ruptures the bond to the adsorbed atom. This surface localization is a two-dimensional extension of that associated with small-polaron formation within solids (Jennison and Emin, 1983b). While small-polaron formation only envisions straining atoms' bonds, desorption also involves bond scission.

The adiabatic approach to self-trapping phenomena, described in Sections 4.1–4.3, is justified when electronic excitations' inter-site transfers are much faster than atoms' vibrations. Thus this approach was also adopted in developing a model of electronically stimulated desorption (Jennison and Emin, 1983b). As illustrated in Fig. 16.1, this desorption process envisions the progressive localization of the electronic excitation with concomitant displacements of adsorbed atoms until an adsorbed atom is expelled as its bond to the semiconductor surface is broken. Two types of desorption

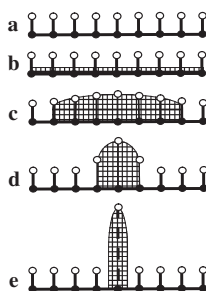


Fig. 16.1 The adiabatic treatment of electronically stimulated desorption is schematically illustrated. Panel *a* depicts absorbed atoms (open circles) bonded to a surface of a crystal's atoms (solid dots). Panel *b* depicts the introduction of a delocalized surface excitation. As atoms' equilibrium positions are progressively displaced (*c* $\rightarrow$ *e*) the electronic excitation's localization increases until the bond to the desorbed atom is finally broken.

can occur. "Immediate desorption" proceeds unimpeded following irradiation. "Delayed desorption" is impeded by the need to traverse an energy barrier.

16.1 Model Hamiltonian

Electronically stimulated atomic desorption from a crystal surface occurs as severe localization of an electronic excitation culminates in breaking an adsorbed atom's bond to the surface. The localization process is analogous to that described in Chapter 4 for the collapse of a charge carrier during small-polaron formation. However, to describe desorption the electron–phonon interaction must include the bond-breaking which occurs as an adsorbed atom is expelled from the semiconductor's surface. Electronically induced bond-breaking is a limit of electronically induced softening. As discussed in Chapter 4, carrier-induced softening is a general feature of polaron formation. Softening occurs because a self-trapped carrier's wavefunction generally changes its size, shape, and centroid as associated atoms vibrate about their displaced equilibrium positions. This effect has frequently been ignored. Indeed, Holstein's molecular-crystal model does not even permit its small polaron to produce carrier-induced softening; its carrier's single-site wavefunction is implicitly represented as just a fixed point (Holstein, 1959a).

In the adiabatic limit atoms' motions become infinitesimally slow as their masses are imagined to become arbitrarily large. The ground-state energy is then just the minimum of the adiabatic potential. The adiabatic potential is the sum of (1) the strain energy associated with displacing adsorbed atoms from their equilibrium positions and (2) the energy of the electronic excitation as a function of these atomic displacements.

Treating the adsorbed atoms bonded to atoms on a covalent semiconductor's surface as a deformable continuum based on elementary structural units of volume V_c , the adiabatic potential becomes

$$V_{\text{ad}} = \frac{k}{2V_c} \int d\mathbf{u} \Delta^2(\mathbf{u}) + \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_r \phi(\mathbf{r})|^2 + \int d\mathbf{r} |\phi(\mathbf{r})|^2 \int d\mathbf{u} \left[F\Delta(\mathbf{u}) - \frac{k'}{2} \Delta^2(\mathbf{u}) \right] \delta(\mathbf{r} - \mathbf{u}). \quad (16.1)$$

The first term in Eq. (16.1) is the Hooke's law strain energy for adsorbed atoms bound to the semiconductor surface with stiffness constant k . Here $\Delta(\mathbf{u})$ denotes the displacement from equilibrium of an adsorbed atom whose nominal location is labeled by $\mathbf{u}$. The second contribution in Eq. (16.1) denotes the kinetic energy of an added electronic excitation having wavefunction $\phi(\mathbf{r})$ and effective mass m . The final contribution in Eq. (16.1) is the expectation value of the electronic excitation's potential as a function of adsorbed atoms' displacements. The first contribution to this term, proportional to $\Delta(\mathbf{u})$, shifts adsorbed atoms' equilibrium positions by amounts proportional to F . The second contribution to this term, proportional to $\Delta^2(\mathbf{u})$, reduces the stiffness with which each adsorbed atom is bound to the semiconductor surface by an amount proportional to k' . The Dirac delta-function $\delta(\mathbf{r} - \mathbf{u})$ expresses the short range of a covalent semiconductor's electron–phonon interaction. With $k' = k$ a severely localized electronic excitation removes the bond between the semiconductor's surface and an adsorbed atom thereby enabling its desorption.

The minimum of the adiabatic potential occurs when its first derivative with respect to $\Delta(\mathbf{u})$ vanishes, cf. Section 4.2:

$$\Delta(\mathbf{u}) = - \frac{FV_c |\phi(\mathbf{u})|^2}{\left[k - k'V_c |\phi(\mathbf{u})|^2 \right]}. \quad (16.2)$$

After some algebraic manipulations the corresponding minimum of the adiabatic potential is found to be

$$V_{\text{ad}}^{\text{min}} = \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_r \phi(\mathbf{r})|^2 - \left(\frac{F^2}{2k} \right) V_c \int d\mathbf{r} \frac{|\phi(\mathbf{r})|^4}{\left[1 - \left(\frac{k'}{k} \right) V_c |\phi(\mathbf{r})|^2 \right]}. \quad (16.3)$$

This result can be compared with Eq. (4.29) with only its short-range electron–phonon interaction. Electronically induced softening of bonds, manifested by $k' > 0$

in the second term of Eq. (16.3), fosters electronic localization by lowering the adiabatic potential's minimum.

16.2 Scaling analysis

As illustrated in Fig. 16.1, desorption results from strain-induced localization of electronic excitations. The scaling approach to polaronic localization detailed in Section 4.3 is an efficient means of studying polaronic localization. In particular, the scaling parameter of this method directly measures polaronic localization. Here the scaling approach of Section 4.3 is applied to Eq. (16.3) in order to study how electronically induced softening affects self-trapping. The electronically stimulated desorption of adsorbed atoms arising from excitation-induced bond-breaking is treated as a special case of electronically induced softening.

The scaling analysis examines the minimum adiabatic energy, Eq. (16.3), as a function of L , a dimensionless scaling factor which alters the electronic excitation's spatial extent (Emin, 1973a; Emin and Holstein, 1976). In particular, the electronic excitation's spatial coordinate $\mathbf{r}$ is replaced by $\mathbf{r}/L$ and its normalized wavefunction is replaced by $L^{-d/2}\phi(\mathbf{r}/L)$ in Eq. (16.3), where d is the electronic wavefunction's dimensionality. Then, upon introducing a change of variable in the integrations of Eq. (16.3), the minimum adiabatic energy is converted to the energy functional

$$V_{\text{ad}}^{\text{min}}(L) \equiv \frac{1}{L^2} \left(\frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_{\mathbf{r}} \phi(\mathbf{r})|^2 \right) - \frac{1}{L^d} \left(\frac{F^2}{2k} \right) V_{\text{c}} \int d\mathbf{r} \frac{|\phi(\mathbf{r})|^4}{\left[1 - \frac{1}{L^d} \left(\frac{k'}{k} \right) V_{\text{c}} |\phi(\mathbf{r})|^2 \right]}. \quad (16.4)$$

Electronically induced softening, having $k' > 0$, lowers the energy functional most severely as L is decreased. Thus, electronically induced softening fosters localization.

For desorption of adsorbed atoms: $d = 2$ and $k' = k$. Then, as illustrated in Fig. 16.2, the energy functional can exhibit two qualitatively distinct behaviors. The energy functional versus L is monotonic when $T_{\text{e}} < V_{\text{s}}$, where:

$$T_{\text{e}} \equiv \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_{\mathbf{r}} \phi(\mathbf{r})|^2 \quad (16.5)$$

is a measure of the electronic kinetic energy and

$$V_{\text{s}} \equiv \left(\frac{F^2}{2k} \right) V_{\text{c}} \int d\mathbf{r} |\phi(\mathbf{r})|^4 \quad (16.6)$$

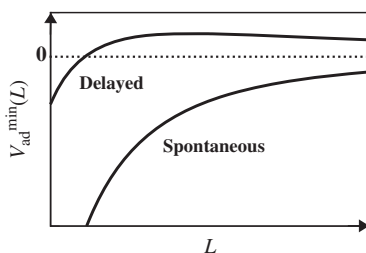


Fig. 16.2 The scaling functional $V_{\text{ad}}^{\text{min}}(L)$ is plotted versus the scaling length L for delayed desorption $T_e > V_S$ and for spontaneous desorption $T_e < V_S$.

is a measure of the short-range electron–phonon interaction. Alternatively, the energy functional is peaked when $T_e > V_S$.

The first instance, $T_e < V_S$, corresponds to “immediate desorption” since an initially delocalized excitation encounters no impediment to achieving full localization and its inducing atomic desorption (Jennison and Emin, 1983b). The second circumstance, $T_e > V_S$, is termed “delayed desorption” since an initially delocalized excitation must traverse an energy barrier before inducing desorption (Jennison and Emin, 1983b).

The preceding discussion has addressed the role that polaronic localization can play in electronically stimulated desorption on an ideal surface. The surface’s reduced dimensionality $d = 2$ and electronically induced softening facilitate the localization required for desorption. As discussed in Section 4.5, surface disorder and traps also generally enhance polaronic localization.

Polaronic phenomena are usually concerned with long-lived carriers associated with atoms’ outer shells. By contrast, electronically stimulated desorption generated by bombardment with very high-energy electrons, photons or ions produces holes within atoms’ inner shells. Electronically stimulated desorption has been attributed to these holes and the associated Auger-generated multi-holes that co-occupy sites (Feibelman and Knotek, 1978; Madden *et al.*, 1982). Desorption is encouraged by the narrow bandwidths associated with such deep-lying multi-holes (Jennison and Emin, 1983b). However, desorption’s efficiency is also limited by the short lifetimes of surface excitations caused by their recombination and diffusion into the solid’s bulk.